CRC
CYCLIC REDUNDANCY CHECK

PRESENTED BY

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INTRODUCTION

A Cyclic Redundancy Check (CRC) in microcontroller programming is an error-detecting code used to spot accidental changes in data during communication or memory storage. It uses a generator polynomial to produce a short checksum value matched by the sender and receiver.

REGISTERS OF CRC

01. Data Register

02. Independent Data Register

03. Control Register

FEATURES

01

Uses CRC-32 (Ethernet) polynomial: 0x04C11DB7

02

Single input/output 32-bit data register

03

CRC computation done in four AHB clock cycles (HCLK)

04

General-purpose 8-bit register

WORK FLOW

AHB (ADVANCED HIGH PERFORMANCE) BUS

DATA REGISTER (INPUT)

CRC COMPUTATION LOGIC

DATA REGISTER (OUTPUT)

POLYNOMIAL EQU.

$$X32 + X26 + X23 + X22 + X16 + X12 + X11 + X10 + X8 + X7 + X5 + X4 + X2 + X + 1$$

THANK YOU VERY MUCH!